PAQ: Prefill-Once, Ask-Many

Reversible, query-conditioned page selection for multi-query document VQA

A living research document — current state, honest negatives, and roadmap

(status: 2026-07-15; this document tracks ongoing work and is updated as results land)

BASED-MM program

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Abstract

Real document-VQA benchmarks are *multi-query*: one long document is prefilled once and asked many different questions (in our pinned evaluation sets, 5.5–5.8 gold-bearing questions per document). Standard KV-cache eviction commits to *one* compressed context and reuses it for every question. We study PAQ ("Prefill-once, Ask-many"): retain the coarse document prefill and, *per query*, re-select a small page subset with the reader's own question→page attention, then reread those pages at full resolution. The axis we test is **reversibility** — per-query re-selection versus commit-once selection — not query-awareness.

What is confirmed. On a 903-query confirmatory run (80 documents per cell, both cells, document-clustered inference), per-query re-selection beats *the identical scorer committed once* (pooled accuracy +0.210, 95% CI $[+0.168, +0.250]$) and a strong gold-informed static subset (+0.106, $[+0.067, +0.141]$) — on both cells, surviving an independent-extension check on the 50 new documents per cell and a worst-case treatment of 4 timeout-missing queries. The same fresh-over-stale pattern also holds for the retrieval scorer tested here (ColQwen2 fresh 0.423 vs. stale 0.178 pooled).

What is equally true and reported with the same prominence. As a page selector, PAQ *loses* head-to-head to a strong trained visual retriever: ColQwen2 re-ranked per query beats PAQ by ≈10 accuracy points at matched page budget (pooled -0.099, 95% CI $[-0.129, -0.072]$), on every reported stratum. Moreover, CAPS (ACL 2026) has already published attention-based page selection beating ColPali-family retrieval on these same benchmarks — the page-level *method* is scooped. A zero-shot expert-head variant (PAQ-T) reaches *recall* parity with ColQwen2 (0.674/0.628 vs. 0.664/0.536 per cell), but its accuracy read-out is still pending, its head is selected using gold recall on complementary document folds, and expert-head selection is CAPS's technique.

What this document is. An honest consolidation of an ongoing project: the confirmed multi-query reversibility result, the first *executed* retrieval-vs-selection panel on MMLongBench-Doc/LongDocURL at 64K context, the evidence that selection quality limits PAQ v1, and the resulting *hypothesis* that selection below whole-page granularity is the next useful direction. The region- and token-level extensions have planning odds of 15–35%, depending on the extension. They are proposals, not results.

Contents

1 Introduction

1.1 The problem

Long-context vision–language models (VLMs) can read an entire multi-page document, but they pay for every page on every question. The standard efficiency response, inherited from single-query text benchmarks, is to *evict*: keep a compact subset of the KV cache (SnapKV, H2O, FastV, LOOK-M, KVzip) and decode against it (Li et al., 2024; Zhang et al., 2023; Chen et al., 2024; Wan et al., 2024; Kim et al., 2025). Eviction is a *single, irreversible commitment*: whatever subset is kept must serve every future question about that document.

1.2 The multi-query regime

Document assistants are not asked one question. The two long-document VQA benchmarks we use are natively multi-query: grouping the 64K splits by source document, MMLongBench-Doc (Ma et al., 2024) averages ≈ 6.97 questions per document and LongDocURL (Deng et al., 2025) ≈ 3.6; inside our pinned evaluation subsets (which require ≥ 3 gold-bearing questions per document) the averages are 5.50 and 5.79 respectively (Table 1). Different questions need different pages, so a subset committed for question 1 is *stale* for question 5.

1.3 The question this project asks

> *At a matched page budget on long multimodal documents, does per-query* ***re-selection*** *of context beat a* ***commit-once*** *selection — and can a training-free, reader-internal selector compete with a dedicated trained retriever?*

The answer, on the evidence in this document, is split, and we state it up front:

- **Yes** to reversibility: re-selection beats commit-once, robustly, on both cells, for our scorer *and* for a retrieval scorer (§7.1, §7.2). This is the confirmed contribution.
- **No** to the training-free selector: PAQ loses to ColQwen2-fresh (Faysse et al., 2025) by $\approx$10 accuracy points at page level (§7.2), and CAPS (Zhu et al., 2026) already published the page-level attention-selection method (§8.1). Zero-shot expert-head scoring closes the *recall* gap (§7.4) but accuracy parity is unverified.
- **Open** below page granularity: our current literature search leaves room for region- and token-level reversible selection. Whole-page routing can spend budget on non-evidence regions, but whether finer selection improves end-to-end accuracy is untested; the roadmap therefore remains explicitly speculative (§9).

1.4 Reading guide

§2 fixes benchmarks and pins; §3 the executed baseline panel; §4 the methods; §5 the statistical machinery; §6 the algorithms; §7 all results (positive and negative, with per-cell results primary and pooled results explicitly labelled); §8 related work including the CAPS overlap; §9 current state and roadmap; §10 limitations. The appendix carries the full panel, the adversarial review trail, and reproducibility pins.

Table 1: The confirmatory evaluation pin (the numbers that should be cited). "Complete queries" are rows where every gated arm scored (complete-case pairing); 4 of 907 vision queries are missing due to a per-document timeout (§7.1).

	LongDocURL	MMLongBench-Doc
Documents	80	80
Complete queries at gate (b=5%)	463	440
Gold-bearing queries / document (pin)	5.79	5.50
Median (mean) union pages / document	44 (45.0)	28 (36.5)
Benchmark-level questions / document (64K split)	≈ 3.6	≈ 6.97

2 Benchmarks and setup

2.1 Cells, documents, pins

We evaluate on two cells: **LongDocURL** (Deng et al., 2025) and **MMLongBench-Doc** (Ma et al., 2024), both consumed as page images (no OCR). For each document we build the *union of the benchmark's own-document pages* across that document's queries (dropping cross-document 64K padding), so every query's evidence is present in one prefill. Documents qualify if they have $\geq$ 3 gold-bearing queries and $\leq$ 110 union pages ($\approx$64K tokens at read resolution); selection is seed-0 frozen.

Two pins exist: the 30-document/cell *screen* pin and the 80-document/cell *confirmatory* pin. The screen pin is a strict prefix of the confirmatory pin (same seed, same arrival order), which is why §7.1 additionally reports an *independent-extension* analysis on the 50 *new* documents per cell — the confirmatory is not independent of the screen unless restricted to those. Both pins are sha-256–checksummed and the harness verifies the screen pin byte-identically before use (Appendix C).

2.2 Reader, budgets, metric

The reader is **Qwen3-VL-8B** (Qwen Team, 2025), frozen, eager attention. All selection arms operate at a matched *page* budget $b \in \{2.5\%, 5\%, 10\%\}$ of the document's pages (the pre-registered gate budget is $b = 5\%$); the selected pages are re-rendered at full resolution and read. Accuracy is the benchmark's document-VQA scorer (exact/relaxed match by answer format). Scope note: everything in this document is a **Qwen3-VL-8B case study**; no second VLM has been run (§10).

2.3 The base-model ceiling

A large fraction of queries receive no credit from this reader even when it is given *exactly* the annotated gold-evidence pages: on the confirmatory run the oracle-evidence arm scores 0 on $317/903 = 35.1\%$ of queries and scores a full 1 on only $407/903 = 45.1\%$ (the pre-specified "answerable subset"). This limits absolute scores. Pairing controls for shared query difficulty, but does not imply that the ceiling affects every selector equally; we report the answerable subset only as a secondary view because it conditions on a model outcome. The oracle row is *not* a strict ceiling: PAQ beats oracle-evidence on 28 queries, because a selected page set can differ from the annotated gold set in ways that help the reader.

Table 2: Arm taxonomy. The load-bearing axis is **reversible** (re-select per query) vs. **commit-once** (one subset per document, reused for all its queries); the second axis is deployability. Disclosed proxies are marked.

Arm	Reversible?	Deployable?	Role
PAQ (ours)	yes	yes	the method
attn_max (ours, max-agg)	yes	yes	reported alternate aggregation
paq_stale	no	yes	**isolation**: same scorer, committed once
gold_frequency_static	no	no (uses gold)	**isolation**: strong gold-informed static
ColQwen2-fresh	yes	yes	**strong deployable retrieval**, re-ranked per query
ColQwen2-stale	no	yes	retrieval committed once (same commit policy)
CLIP-fresh	yes	yes	cheaper retrieval control (Radford et al., 2021)
SnapKV-fresh	yes	yes	reference (weak port; beating it not required)
SnapKV-stale	no	yes	strongest stale family arm
H2O / FastV / LOOK-M / KVzip	no	yes	demoted query-blind controls (proxies, see text)
random / truncation	no	—	controls
full-context / oracle-evidence	—	—	reference rows, *not* competitors

3 Baselines: the executed panel

Every arm below was actually run, per query, at matched page budget, on the same pinned data with the same reader. The resulting panel directly compares retrieval and reader-internal selection on MMLongBench-Doc/LongDocURL at 64K in the multi-query regime; our current literature search did not identify an earlier executed panel with this exact scope. Table 2 gives the taxonomy; Appendix A the full numbers.

The demoted eviction family (disclosed proxies). H2O uses accumulated received attention; FastV is a *context-only proxy* (canonical FastV is prompt-conditioned); LOOK-M degenerates to H2O on image-only pages; KVzip is a `recv_max` proxy for the reconstruction pass of Kim et al. (2025). On these cells the whole family sits *at or below random* page recall, so beating it is not a claim we make — they are query-blind controls, retained for completeness, and a faithful-reconstruction KVzip would likely be stronger. The honest comparators are `paq_stale`, `gold_frequency_static`, SnapKV-stale/fresh, and the retrieval arms.

On `gold_frequency_static`. This arm keeps the top-B pages by *gold-page frequency* across the document's queries. It uses gold labels, so it is not deployable; it is a *strong gold-informed static commitment* — but it is **not** provably accuracy-optimal and we do not call it "best possible" or a "ceiling" (an earlier draft did; the claim was retracted under review, Appendix B).

Reference rows. Full-context and oracle-evidence are reported for calibration only. SnapKV-fresh (re-run per query) is a reference, not an upper bound; our port averages heads and layers before pooling and is weaker than canonical SnapKV.

4 Methods

4.1 PAQ: coarse-to-fine page routing

PAQ is training-free page routing over a frozen VLM:

1. **Prefill once (per document).** Pages are rendered at coarse resolution ($\approx$3.5K visual tokens total, i.e. $\approx$70 tokens/page on a 50-page document) and prefilled once; the KV cache is retained. Attention statistics are captured by OOM-safe per-layer forward hooks (Algorithm 4).

2. **Re-select per query.** The question is appended to the retained cache; the question rows' attention over page tokens (mean over heads, layers, and question tokens) is averaged over each page's token span to give a per-page score (Algorithm 1). Keep the top pages at budget b.

3. **Reread.** The selected pages are re-rendered at full resolution and the question is answered from them.

Scope, stated plainly: this is *page routing* — coarse-score → pick pages → full-resolution reread. It is *not* token-level KV compression: we never decode from a masked retained cache, and we claim no memory savings in v1. The estimand is query-conditioned page selection at matched page budget.

4.2 The reversibility framing and the isolation protocol

"Query-aware selection beats query-agnostic eviction" is both a known text result (Tang et al., 2024) and a strawman risk (FastV and LOOK-M can attend to the question). We therefore fix the axis to **reversibility**: every baseline runs its canonical scorer but is deployed *stale* — committed once per document (query-aware members commit an aggregate over the document's first $M = \min(3, k)$ queries) — while PAQ re-selects per query (Algorithm 2).

Reversibility is credited *only* if PAQ beats **both**

- `paq_stale`: PAQ's *own* scorer committed once. Identical scorer, identical budget; the only difference is reversibility, so the delta isolates the mechanism;

- `gold_frequency_static`: the strong gold-informed static, so the win is not an artifact of a weak committed selector.

This gate exists because a first, weaker gate (PAQ vs. best-of-eviction-family) *passed and was then retracted*: adversarial review showed the family's stale arms were weak-static artifacts (a gold-informed static nearly matched PAQ in recall), so "reversibility" had not been isolated. The two-comparator gate above was frozen before the confirmatory run (Appendix B documents both review rounds).

4.3 PAQ-T: expert-head, higher-resolution scoring (training-free CV)

After the retrieval loss (§7.2) we tested how much of the gap is an artifact of PAQ's mean-aggregation and resolution starvation ($\approx$70 coarse tokens/page vs. ColQwen2's $\approx$1,024 patch embeddings/page). PAQ-T scores with (a) more scoring tokens per page (256 tokens/page, 6,000-token target total — a cap forced by encoder OOM on large documents, not full page resolution), and (b) a single *expert (layer, head)* chosen by held-out-document cross-validation (Algorithm 3) rather than the mean over all heads. Expert-head selection is CAPS's technique (Zhu et al., 2026); what is ours here is the amortized multi-query architecture it plugs into. PAQ-T so far has a *recall* result only (§7.4); the accuracy read-eval is pending. "Training-free" means no parameter updates: choosing the head still uses gold recall on the complementary fold of the same benchmark, so this is not unsupervised model selection.

5 Definitions and statistical machinery

Matched page budget. For a document with n pages and budget fraction b, every arm keeps

$$B \;=\; \max\bigl(1, \lfloor b \cdot n \rfloor\bigr) \tag{1}$$

pages. Pages have unit budget weight in the executed vision harness, so this is the exact behavior of its cumulative-budget selector; ties in score are broken by page index. The same selector code path is used for every arm.

Gold-page recall at budget. For query q with nonempty annotated gold-page set G_q and kept set K_q ($|K_q| = B$),

$$\text{recall@}b(q) \;=\; \frac{|K_q \cap G_q|}{|G_q|}. \tag{2}$$

Reported recall is the query-macro mean of this fraction within a cell; it is not answer accuracy and it is not the binary event "at least one gold page retrieved."

Commitment pressure. For a document with g distinct gold pages across its k queries,

$$\text{pressure} \;=\; \frac{g}{b \cdot n}, \tag{3}$$

i.e. how many times over the b-budget capacity the document's queries collectively need. This pre-registered quantity uses the fractional, pre-rounding capacity bn, whereas the executed selector uses B from Eq. (1). Thus pressure > 1 means $g > bn$; when $bn \geq 1$ this also implies $g > B$ and no single B-page subset can contain every distinct annotated gold page. For $bn < 1$ and at rounding boundaries, pressure should be read as a continuous diagnostic rather than a literal coverability test. This was the pre-registered mechanism axis; the formal test of it **failed** (§7.3) and it is retained as exploratory only.

Document-cluster paired bootstrap and LB_{95}**.** For arms A and C, let $d_q = s_A(q) - s_C(q)$ be their paired accuracy-score difference on query q. With two cells $c \in \{1, 2\}$ and N_c queries in cell c, the pooled point estimate is

$$\widehat{\Delta}_{A-C} = \frac{1}{2} \sum_{c=1}^{2} \left(\frac{1}{N_c} \sum_{q \in c} d_q \right). \tag{4}$$

Queries of one document share the document *and* the single stale subset, so bootstrapping over queries would be anti-conservative pseudoreplication. All intervals therefore resample *documents* with replacement within each cell (10,000 draws, seed 0), recompute Eq. (4), and take percentile endpoints. Per-cell intervals use the same procedure with one cell. We define

$$\mathrm{LB}_{95}(\Delta) = 2.5\text{th percentile of the bootstrap distribution of } \Delta, \qquad \mathrm{UB}_{95}(\Delta) = 97.5\text{th percentile}, \tag{5}$$

i.e. LB_{95} is the **lower end of a two-sided 95% CI** (equivalently a one-sided 97.5% bound). An earlier label of "one-sided 95%" was incorrect and has been fixed; the frozen gate arithmetic never changed.

Algorithm 1 PAQ per-query scorer and matched-budget selection (vision cells)

Require: document pages $P_1, \ldots, P_n$; queries $q_1, \ldots, q_k$; budget b; frozen VLM (eager attention)

render all pages coarsely ($\approx$3.5K visual tokens total); prefill once; retain KV cache C
record image-token spans $s_1, \ldots, s_n$ for the pages
for each query q_j **do**
    forward only the tokenized question against C (no re-prefill)
    via hooks: $\mathbf{a} \leftarrow$ mean over layers, heads, and question rows of attention to each context position ▷ Algorithm 4
    per-page score $\text{score}_j(p) \leftarrow |s_p|^{-1} \sum_{t \in s_p} \mathbf{a}[t]$ ▷ `attn_max` variant: max over s_p
    $K_j \leftarrow$ top-B pages by score_j, $B = \max(1, \lfloor b\, n \rfloor)$ ▷ Eq. (1); index tie-break
    re-render pages K_j at full resolution; answer q_j from them
end for

Algorithm 2 Commit-once (stale) vs. re-select (fresh) deployments — the reversibility contrast

Require: per-query score vectors $v_1..v_k \in \mathbb{R}^n$ from ANY scorer (PAQ, SnapKV, ColQwen2)

fresh (reversible): for query q_j, select top-B from v_j ▷ one subset per query
stale (commit-once): $u \leftarrow \frac{1}{M} \sum_{j \le M} \text{minmax}(v_j)$ with $M = \min(3, k)$; select top-B from u once; reuse that subset for ALL k queries
min-max normalization before averaging prevents any single query's magnitude from dominating the committed ranking; query order is frozen (arrival order).
query-agnostic arms (H2O/FastV/LOOK-M/KVzip/random/trunc) are stale by construction (their score vector never sees a question).

The isolation gate (pre-registered, frozen before the run).

$$\textsc{Isolated} \iff \text{LB}_{95}\big(\widehat{\Delta}_{\text{PAQ}-\texttt{paq\_stale}}\big) > 0 \;\; \wedge \;\; \text{LB}_{95}\big(\widehat{\Delta}_{\text{PAQ}-\texttt{gold\_frequency\_static}}\big) > 0, \tag{6}$$

using complete cases at $b = 5\%$. The frozen gate is pooled with equal cell weighting; the per-cell intervals are corroborating analyses, not additional gate clauses. If either interval includes zero, the evidence is insufficient to isolate a reversibility benefit; that outcome would not establish equivalence between commit-once and per-query selection.

6 Algorithms

7 Results

The primary confirmatory numbers below come from `research/paq_verdict.json` (903 complete vision queries, 80 documents/cell, frozen v3 harness). The independent-extension, mechanism, and fusion diagnostics come from the documented post-hoc scripts in Appendix C; PAQ-T numbers come from `research/paq_t_gate_verdict.json`. Per-cell results are primary. "Pooled" inferential contrasts use the equal-cell statistic in Eq. (4); query-micro summaries are labelled separately.

7.1 Confirmatory isolation: reversibility is real

CONFIRMED Per-query re-selection beats the identical scorer committed once AND a strong gold-informed static, on both cells, document-clustered, surviving independent-extension and missingness worst-case checks.

Algorithm 3 Expert-head selection with held-out-document CV (PAQ-T; gold-supervised selection on the complementary fold, never on the scored documents)

Require: per-(layer,head) question→page scores; gold sets G_q; documents D of one cell; budget $b = 5\%$

1: split D into two folds by document (seed-0 shuffle)

2: **for** each fold F **do**

3: $\quad Q_{\text{train}} \leftarrow$ queries in $D \setminus F$

4: $\quad$ on Q_{train}: pick $(\ell^*, h^*) = \arg\max_{\ell,h} |Q_{\text{train}}|^{-1} \sum_{q \in Q_{\text{train}}} \text{recall@}b_{\ell,h}(q)$ $\triangleright$ Eq. (2)

5: $\quad$ evaluate (ℓ^*, h^*) on the held-out fold F

6: **end for**

7: report query-macro held-out recall per cell $\triangleright$ no weight updates; head choice does use training-fold gold labels

Algorithm 4 OOM-safe per-layer-hook attention capture (the engineering that makes 64K scoring feasible)

1: register a forward hook on every language-decoder self-attention module (vision encoder excluded)

2: run the chunked prefill / question forward with `output_attentions=False` under eager attention

3: inside each hook: catch the layer's $(1, H, q, k)$ attention tensor, reduce it immediately (fp32-accumulated column-sum for H2O, running max for KVzip, layer-2 last row for FastV, question-row mean for PAQ), and **discard** it

4: peak memory = ONE layer's attention. The naive all-layers `output_attentions=True` path OOMs (≈77 GB at 128K text) or crashes vision cuDNN.

5: a capture guard hard-fails if no hook observed a 4D tensor (prevents silently scoring zeros)

Table 3 and Figure 1 give the gate result. At $b = 5\%$:

At screen scale (30 documents/cell), the MMLongBench-Doc contrast against the static was -0.017 with a CI including zero. It became positive at 80 documents; we treat that earlier result as an underpowered screen, not as an independent replication.

Independent extension (the screen is embedded in the confirmatory). Because the 80-document pin contains the 30 screen documents, we re-ran the isolation statistics on the **50 new documents per cell only** ($n = 582$ queries): PAQ − `paq_stale` +0.216 $[+0.159, +0.269]$; PAQ − `gold_frequency_static` +0.108 $[+0.059, +0.150]$; per-cell static contrasts +0.089 $[+0.004, +0.162]$ (LongDocURL) and +0.127 $[+0.083, +0.173]$ (MMLongBench-Doc). The result therefore also holds on the out-of-screen extension.

Answerable subset (secondary). On queries where oracle-evidence scores exactly 1 ($n = 407$, 137 docs): PAQ 0.583 vs. `paq_stale` 0.182 (+0.395 $[+0.324, +0.460]$) and vs. the static 0.382 (+0.204 $[+0.135, +0.266]$). Reported secondarily because it conditions on a model outcome.

Missingness. 4 of 907 vision queries (MMLongBench-Doc document `mi_phone`, queries 2–5) are missing due to a per-document timeout. Imputing them worst-case in either direction leaves the gate verdict unchanged.

Table 3: Confirmatory isolation at b=5% on 903 complete queries (accuracy; document-cluster paired bootstrap). Contrast rows report $\Delta\,[\mathrm{LB}_{95}, \mathrm{UB}_{95}]$; the pooled column uses equal cell weighting.

	LongDocURL	MMLongBench-Doc	pooled (equal-cell)
PAQ accuracy	0.385	0.258	0.322
`paq_stale` accuracy	0.128	0.095	0.111
`gold_frequency_static` accuracy	0.278	0.154	0.216
PAQ − `paq_stale`	+0.257 [+0.184, +0.330]	+0.163 [+0.126, +0.200]	+0.210 [+0.168, +0.250]
PAQ − `gold_freq_static`	+0.108 [+0.045, +0.164]	+0.104 [+0.063, +0.144]	+0.106 [+0.067, +0.141]
Verdict	Reversibility isolated (both $\mathrm{LB}_{95} > 0$, both cells and pooled)		

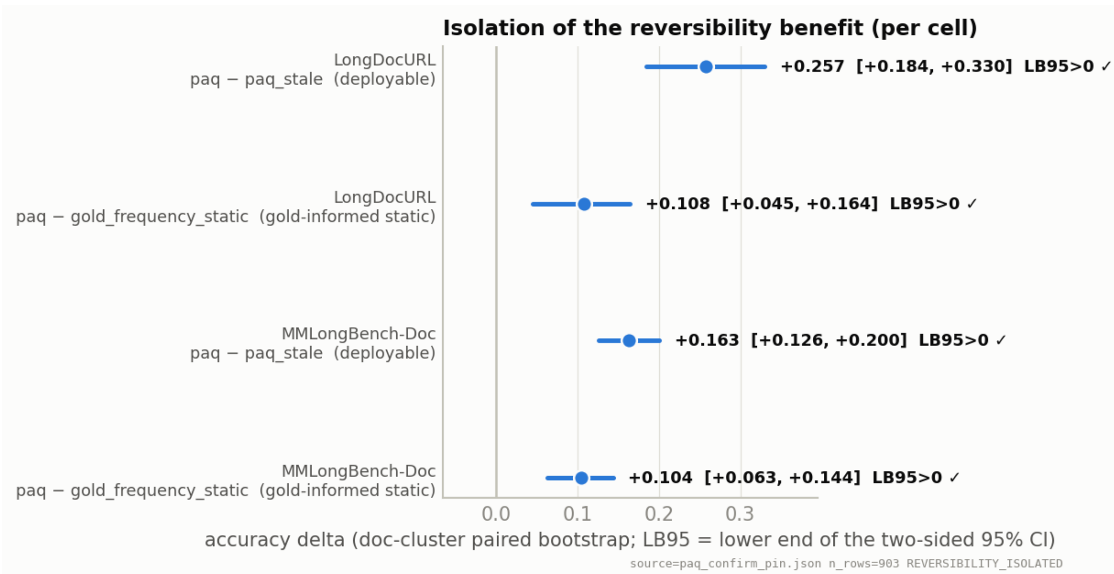

Figure 1: The statistical headline: per-cell forest of the two isolation contrasts at $b = 5\%$. Intervals are document-clustered two-sided 95% CIs; all four exclude zero.

Budget curves. Figure 2 shows accuracy at $b \in \{2.5, 5, 10\}\%$: PAQ dominates `paq_stale` and the static at every budget on both cells, monotonically. On LongDocURL, PAQ at $b = 10\%$ reaches 0.518 vs. full-context 0.546 — reading 10% of pages recovers ≈95% of full-context accuracy on that cell (not on MMLongBench-Doc: 0.317 vs. 0.397).

7.2 The deployable retrieval comparison: PAQ loses at page level

CONFIRMED NEGATIVE A strong trained visual retriever re-ranked per query beats PAQ by ≈10 accuracy points at matched page budget. This is the decisive result of the final review round, reported with the same prominence as the positive.

The adversarial AAAI-readiness review named one decisive missing experiment: a deployable query-conditioned *retrieval* baseline. We ran it: ColQwen2 (Faysse et al., 2025) (late-interaction MaxSim over page images; pages embedded once per document) fresh and stale, plus CLIP (Radford et al., 2021) fresh, same reader, same budgets, all 903 queries. At $b = 5\%$, pooled, document-clustered (Table 4, Figure 3):

Three readings, all of which we stand behind:

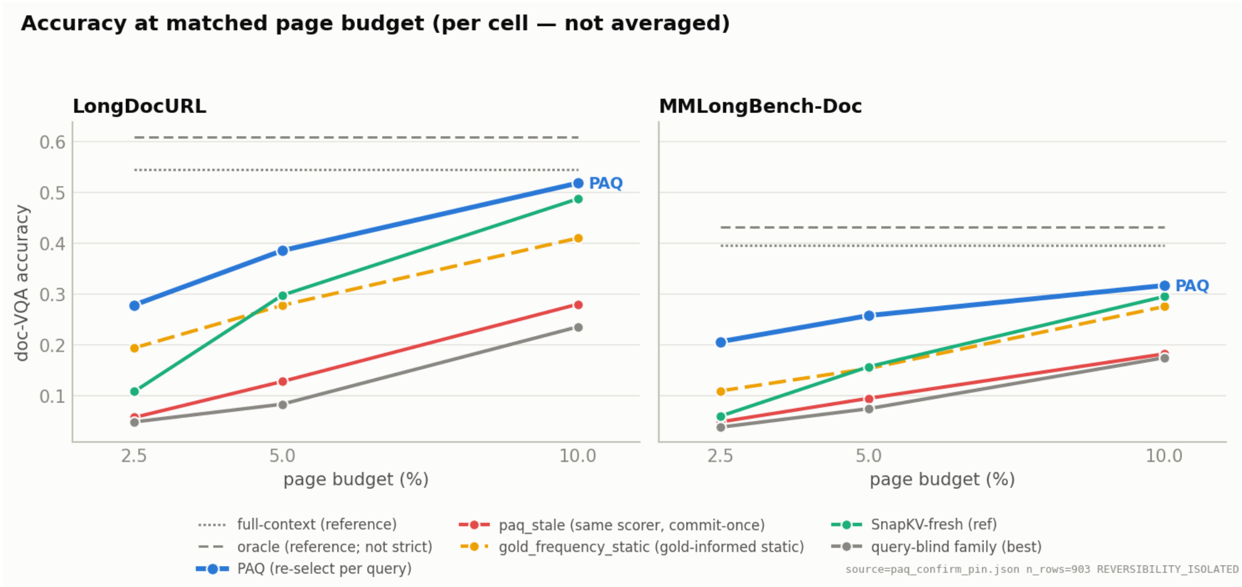

Figure 2: Accuracy at matched page budget, per cell (never averaged). Full-context and oracle-evidence are recessive reference lines, not competitors; the "query-blind family (best)" line is the demoted control family's best member.

1. **PAQ is not a superior page router.** The loss is ≈10 points, LB_{95} well below zero, and it holds on every stratum we examined (single-gold −0.110, multi-gold −0.086; no question type where attention beats retrieval).

2. **The reversibility pattern spans both tested scorer families.** On the query-micro summary, ColQwen2-fresh is 0.423 vs. 0.178 stale; the corresponding equal-cell means are 0.420 and 0.177. This reproduces the fresh-vs-stale collapse with a different scorer. It supports per-query reselection for attention- and retrieval-based systems in this evaluated regime; it is not a claim about every possible scorer.

3. **The two existing outputs have limited complementarity.** A per-query oracle taking the better observed answer score from {PAQ, ColQwen2-fresh} gains +0.035 pooled over ColQwen2-fresh (LB_{95} +0.024; +0.041/+0.030 per cell). Among the 419 queries where ColQwen2-fresh scores exactly zero, PAQ scores one on only ≈6%. This makes simple output routing low priority. It does *not* upper-bound a fusion or reranker that selects a new page set, whose reader output could differ from both inputs.

Gold-page recall is consistent with a coverage contribution to the loss: at $b = 5\%$, ColQwen2-fresh recalls 0.664/0.536 (per cell) vs. PAQ 0.494/0.443. This observation motivated the PAQ-T selection diagnostic (§7.4); by itself it is not a causal decomposition of the accuracy gap.

Why the loss redirects rather than ends the project. Simply replacing mean aggregation with a better page-level attention head is not a clean novelty route: CAPS has already published expert-head attention selection on these benchmarks (§8.1). We therefore treat PAQ-T as a diagnostic and defer the unexecuted below-page ideas to §9, after the confirmed and negative evidence.

Table 4: Deployable retrieval comparison at b=5% (accuracy; 903 complete queries). PAQ loses to ColQwen2-fresh on both cells. Fresh exceeds stale for both tested scorers. The pooled arm means use equal cell weighting; contrast intervals are document-clustered.

	LongDocURL	MMLongBench-Doc	pooled (equal-cell)
ColQwen2-fresh	0.512	0.328	0.420
PAQ	0.385	0.258	0.322
ColQwen2-stale (commit-once)	0.216	0.138	0.177
CLIP-fresh	0.155	0.164	0.159
full-context (reference)	0.546	0.397	—
PAQ − ColQwen2-fresh	pooled -0.099 $[-0.129, -0.072]$ $\Rightarrow$ **PAQ loses**		
PAQ − ColQwen2-stale	pooled $+0.145$ $[+0.101, +0.185]$		
PAQ − CLIP-fresh	pooled $+0.162$ $[+0.128, +0.197]$		

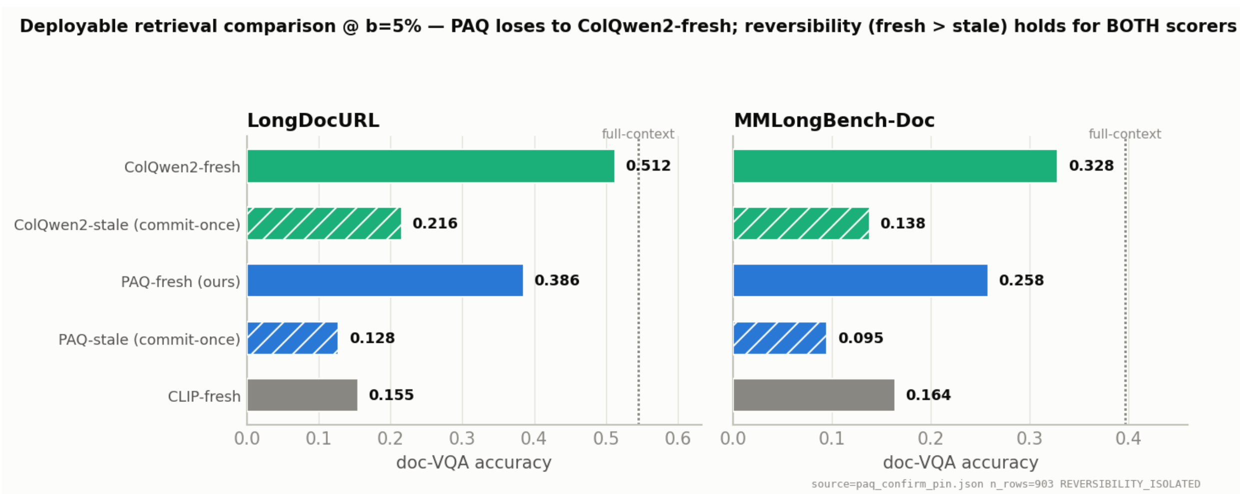

Figure 3: The two honest facts in one picture, per cell: (i) ColQwen2-fresh beats PAQ (the ≈10pt loss); (ii) *both* scorers collapse when committed once — reversibility is a general property of the multi-query regime, not a PAQ-specific effect.

7.3 Mechanism: demoted to exploratory

RETRACTED AS A CLAIM The pre-registered "commitment-pressure gradient" mechanism FAILED its formal test; what replaced it is a corrected, two-part account of where the benefit comes from.

We had hypothesized the reversibility benefit concentrates in high-commitment-pressure documents (Eq. (3)) and reverses under low pressure. The screen suggested this; the formal document-clustered interaction test on the confirmatory does **not** support it: high−low pressure difference $+0.042$ $[-0.048, +0.130]$ on PAQ−static and -0.015 $[-0.116, +0.080]$ on PAQ−`paq_stale` — both include zero. LongDocURL shows the predicted direction without significance; MMLongBench-Doc's benefit is broad across pressure. Figure 4 is therefore labelled exploratory and this document makes no mechanism-gradient claim.

An earlier "reading quality, not coverage" explanation was also **wrong**: PAQ's gold-recall exceeds the static's by $+0.092$ (the earlier "recall ≈ static" countercheck was in error), and at *matched* recall ($n = 415$) PAQ still beats the static by $+0.049$ $[+0.021, +0.076]$. The corrected descriptive statement is: **per-query re-selection has higher gold coverage overall and higher down-**

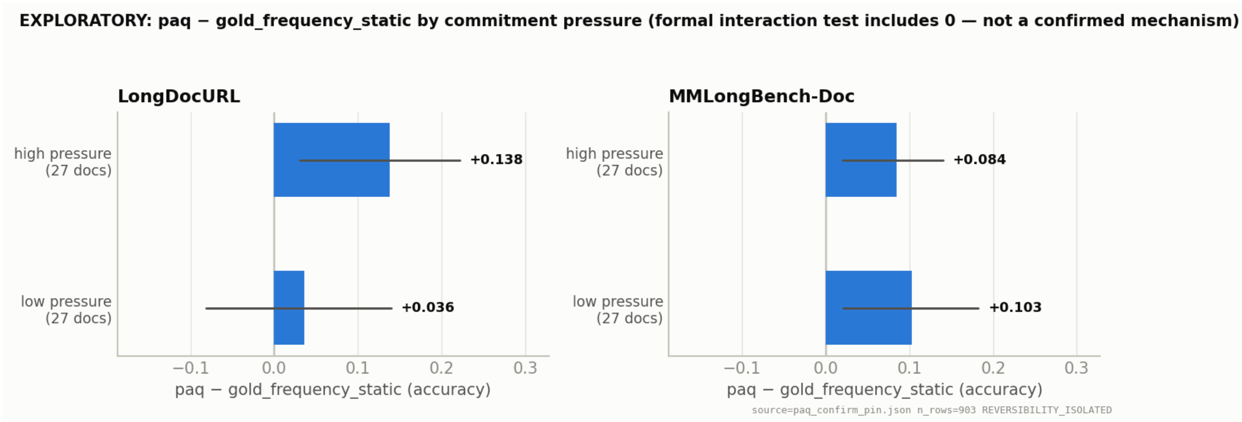

Figure 4: EXPLORATORY (demoted): PAQ−`gold_frequency_static` by commitment-pressure split. The formal high−low interaction includes zero on both contrasts; shown for completeness, not as a confirmed mechanism.

Table 5: PAQ-T page-selection recall@b=5% diagnostic. Sample sizes differ, so rows should not be read as paired deltas. **This is recall, not answer accuracy**; the read-eval is pending.

Scorer	LongDocURL	MMLongBench-Doc
coarse mean aggregation (PAQ v1)	0.494 ($n = 463$)	0.443 ($n = 440$)
higher-resolution mean aggregation	0.523 ($n = 451$)	0.460 ($n = 425$)
expert head (held-out-document CV)	0.674 ($n = 451$)	0.628 ($n = 425$)
ColQwen2 in-harness reference	0.664 ($n = 463$)	0.536 ($n = 440$)

stream accuracy among queries with equal measured gold recall. The matched-recall analysis does not identify why the selected page sets differ or establish a separate causal "reading quality" mechanism.

7.4 PAQ-T: training-free CV recall result — accuracy pending

PARTIAL / PENDING Expert-head + higher-resolution scoring descriptively meets or exceeds the ColQwen2 RECALL means. This is neither a paired non-inferiority result nor accuracy: the read-eval has not been run, and the expert-head technique is CAPS's.

The PAQ-T gate (score-only; no reads) asks whether the recall deficit behind the −10pt loss is an artifact of mean aggregation and scoring resolution. Its predeclared engineering verdict is `PAQ_T_VIABLE`: the worst displayed per-cell gap to ColQwen2 is −0.96pt, inside the 10-point viability threshold. Held-out-document expert-head recall@b=5% is **0.674** (LongDocURL, $n = 451$) and **0.628** (MMLongBench-Doc, $n = 425$). ColQwen2's in-harness means are 0.664 and 0.536 (reference-pipeline values 0.694/0.555).

These are *not* paired estimates on identical rows: PAQ-T covers 876 high-resolution-score rows, whereas the coarse-PAQ and ColQwen2 means use all 903 complete rows. Accordingly, Table 5 is a descriptive diagnostic, not a confidence-interval-backed parity test. Within the displayed sequence, the largest change is from high-resolution mean aggregation (0.523/0.460) to the cross-validated expert head (0.674/0.628); the coarse-to-high-resolution increment cannot be cleanly attributed without recomputing both on the same 876 rows. Selected heads lie in layers 18–21 (head $(21, 10)$ is selected in one fold of each cell).

Four caveats are load-bearing: (1) **recall parity ≠ accuracy parity** — the read-eval (reread expert-head top-B pages, score against ColQwen2) is the required next experiment, and until it runs PAQ-T has no accuracy claim; (2) **expert-head selection is gold-supervised cross-validation and CAPS's method** (Zhu et al., 2026) — no model weights are trained, but gold recall on the complementary benchmark fold chooses the head, and our residual novelty is the amortized one-prefill-many-queries architecture rather than the selector; (3) the available-case mismatch above prevents an inferential non-inferiority claim; and (4) scoring resolution was capped by encoder OOM (6,000-token target), so the resolution lever remains under-tested. The result supports running the read-eval; it does not establish page-router parity.

7.5 Efficiency framing (measurement pending)

UNMEASURED No latency/VRAM/throughput table exists yet. Everything in this subsection is an accounting argument, flagged as such, with the red-team's caveats folded in.

PAQ is training-free, uses no separate retriever model and no multi-vector index; the coarse prefill is paid once per document and each query costs one short forward over the retained cache plus the reread. But the honest accounting (from the efficiency red-team) is: ColQwen2 *also* amortizes (page embeddings are computed once; a query costs a small query encoding plus MaxSim), and PAQ's per-query readout is a reader-scale forward — likely *more* per-query FLOPs than ColQwen2's query encoding. What PAQ genuinely saves is *dependencies*: no second model resident, no index storage, one deployed artifact. Where the amortization argument has real teeth is against CAPS, which re-forwards every page per query (11.25 s/query reported) and explicitly concedes the multi-query regime to embedding methods. An efficiency-only justification while losing ≈10 accuracy points was reviewed as weak-reject; we therefore make no efficiency claim until (a) the measured table exists (cold/warm start, retriever co-resident vs. sequential, index bytes/page, several queries-per-document), and (b) PAQ-T accuracy lands within the predeclared non-inferiority bars (within 2–3 points, $\mathrm{LB}_{95} > -0.03$, both cells).

8 Related work

8.1 CAPS: the page-level method is already published

CAPS (Zhu et al., 2026) ("Attention as Selector", ACL 2026) publishes attention-based page selection for long-document retrieval *on our exact benchmarks*: a small scoring VLM's final-token→page attention at a single expert (layer, head), lightly calibrated on 6.4k off-benchmark contrastive triplets, beats ColPali-pipeline retrieval (MMLongBench R@5 78.89 vs. 72.00; downstream QA 44.69 vs. 36.18–41.01). Their training-free ablation lands just *below* ColPali (best zero-shot head R@5 70.92 vs. 72.00) — independently corroborating our own training-free loss. We state plainly: **the "reader-attention selects pages" method is scooped**, and any contribution of ours must be argued elsewhere. CAPS's own concessions define the open ground: it scores every page independently per query (multi-query latency conceded to embedding methods — exactly PAQ's amortized regime) and operates at page granularity only (region-level selection named as an open problem — exactly extension E1).

8.2 KV compression and eviction

H2O (Zhang et al., 2023), SnapKV (Li et al., 2024), FastV (Chen et al., 2024), LOOK-M (Wan et al., 2024), and KVzip (Kim et al., 2025) commit a compressed cache once; Quest (Tang et al.,

2024) shows query-aware beats query-agnostic at tight budgets in single-query text — motivation for, but not evidence of, our multi-query claims. SparseVILA (Anonymous, 2025d) is the nearest VLM neighbor for extension E2 (prefill pruning + decode-time retrieval) but prunes irreversibly at prefill and never compares against retrieval pipelines on long-document VQA; Kamera (Anonymous, 2026c) shows the multi-query KV-reuse serving regime is becoming real.

8.3 Trained visual retrieval

ColPali/ColQwen2 (Faysse et al., 2025) is the strong deployable retriever we lose to; M3DocRAG (Cho et al., 2024) and successors are the pipelines CAPS beat. The bar is moving: ColQwen3-class models and small distilled retrievers keep improving, so even a future win over ColQwen2 must be argued as a mechanism/regime result, not a leaderboard entry. A published oracle-analysis on MMLongBench (Anonymous, 2026b) mirrors our fusion finding (best-per-query routing between retrievers gains +0.9; union reranking gains more), and attention-based reranking precedents exist in text (Chen, Gutierrez, and Su, 2025; Wu et al., 2025).

8.4 Benchmarks and the region-level neighborhood

MMLongBench-Doc (Ma et al., 2024) and LongDocURL (Deng et al., 2025) supply the multi-query long-document structure; MMDocIR (Dong et al., 2025) adds layout-level retrieval labels useful for E1's localization sanity checks. The region-level neighborhood is active — RegionRAG (Yuan et al., 2025) (trained retriever-side crops; near-flat on long docs, no long-doc QA), Snappy (Anonymous, 2025c) (training-free patch→OCR-box propagation, localization only), AGREE (Anonymous, 2025a) (reader attention distilled into a trained retriever), VRAG-RL (Wang et al., 2025) and Doc-V* (Anonymous, 2026a) (agentic crop/zoom loops), ViCrop/FOCUS (Zhang et al., 2025; Anonymous, 2025b) (single-image training-free cropping). In our current search we did **not identify prior work in the exact combined cell**: reader-internal attention as the runtime region selector, end-to-end QA at matched token budget vs. page-level retrieval, on long documents, multi-query amortized. This is a provisional literature-map claim, not proof of novelty; it motivates the E1 proposal. Page-selection-then-read itself long predates us (Tito, Karatzas, and Valveny, 2024).

9 Current state, open questions, roadmap

9.1 Scoreboard (2026-07-15)

Item	Status	Where
Multi-query reversibility isolation (page level)	**CONFIRMED** (903 q, both cells)	§7.1
Executed retrieval-vs-selection panel at 64K	**DONE** (first of its kind, to our knowledge)	§3, App. A
PAQ vs. ColQwen2-fresh	**LOST** (-0.099 pooled)	§7.2
Page-level attention selection as a method	**SCOOPED** (CAPS, ACL 2026)	§8.1
Commitment-pressure mechanism	**DEMOTED** (interaction test incl. 0)	§7.3
PAQ-T zero-shot recall parity	**CONFIRMED (recall only)**	§7.4
PAQ-T accuracy read-eval	**PENDING** (the next experiment)	§7.4
Efficiency table	**PENDING** (unmeasured)	§7.5
E2 PAQ-KV (token-level reversible KV)	**ROADMAP**, odds ∼25–35%	§??
E1 PAQ-Fovea (region-level foveation)	**ROADMAP**, odds ∼15–25%	§??

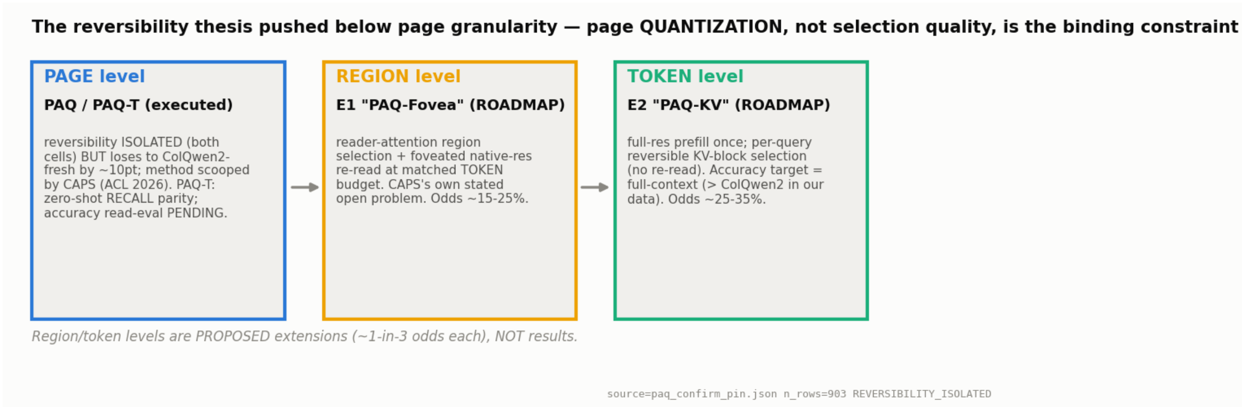

Figure 5: The page → region → token reversibility roadmap. Page level is executed (and lost/scooped); region and token levels are *proposals* at ∼1-in-3 odds each. The unifying observation: retrieval quantizes evidence to whole pages, so at a fixed budget the open opportunity is below page granularity, where a reader that has already prefilled the document can re-select reversibly per query.

9.2 The thesis going forward

The reframe that survived every review round: **page quantization, not selection quality, is the binding constraint**. ColQwen2 wins the page-ranking game, and CAPS owns the attention variant of it — but a whole-page budget is wasteful (most of a kept page is not evidence), and a page-level index cannot re-allocate budget within pages. The reader's retained prefill can. The two proposed operating points (Figure 5):

- **E2 / PAQ-KV (token):** the best-priced bet — its accuracy target (full-context) already beats ColQwen2-fresh in our own data (+0.033/+0.069), so it needs selection loss $<\sim$1pt rather than a swing. Risks: a 2–4 day engineering spike (Qwen3-VL block-top-k under M-RoPE/DeepStack), and the LongDocURL margin (+0.033) is inside our CI half-width — a per-cell $\mathrm{LB}_{95} > 0$ there requires landing essentially at full-context.
- **E1 / PAQ-Fovea (region):** the highest-novelty bet (CAPS's own named open problem; the cell is unoccupied, §8); needs the largest accuracy swing (∼+0.14 total) and its localization sanity at coarse resolution is unproven.

Both are ∼**1-in-3 propositions and are stated as such**. If neither survives its cheap screen, the honest terminus of this program is what is already in this document: the reversibility regime result, the executed panel, and the boundary map.

9.3 Open questions

(1) Does PAQ-T recall parity convert to accuracy non-inferiority (the pending read-eval)? (2) Does the E2 spike land, and does token-level selection preserve full-context accuracy on multi-hop/cross-page questions (Quest-family methods degrade exactly there, and MMLongBench-Doc is 33% cross-page)? (3) Can coarse-prefill heatmaps localize regions at all (E1's gate)? (4) The moving bar: ColQwen3-class retrievers may outdate any ColQwen2 comparison by submission time — any win must be a regime/mechanism claim. (5) A second VLM (currently everything is a Qwen3-VL-8B case study).

10 Limitations

- **Heterogeneity.** Effect sizes differ by cell (e.g. the stale gap is +0.257 vs. +0.163); we never average cells except in the labelled pooled statistic, and per-cell results are primary.
- **Page routing only.** v1 selects pages and rereads at full resolution; it is not KV compression, makes no memory/latency claim, and the systems-level story is unmeasured (§7.5).
- **Base-model ceiling.** 35.1% of confirmatory queries are oracle-unanswerable (score 0 given exactly the gold pages); only 45.1% are fully oracle-answerable. Absolute numbers are deflated for every arm; oracle is not a strict ceiling (28 PAQ wins over it).
- **Resolution cap.** The high-res vision encoder OOMs on many-page documents, so PAQ-T scoring was capped at a 6,000-token target — the resolution lever is under-tested and a PAQ-T-negative would have been inconclusive.
- **CAPS overlap.** Expert-head attention selection is CAPS's published method; any PAQ-T claim is limited to the amortized multi-query architecture and must credit CAPS.
- **Recall $\neq$ accuracy.** The only PAQ-T evidence is recall; the accuracy read-eval is pending and could fail.
- **Weak family ports.** FastV/LOOK-M/KVzip are disclosed proxies and SnapKV is a weak port; we claim nothing against canonical versions of them.
- **Single reader.** All results are a Qwen3-VL-8B case study; generalization to other VLMs is untested.
- **Stale-policy dependence.** The commit-once aggregate uses the first $M = \min(3, k)$ queries in frozen arrival order; order-permutation and M-sensitivity robustness checks were reviewer-requested and remain to be run (the gold-informed static, which is order-free, partially covers this).

A Full executed panel at the gate budget

Secondary statistics (pooled, reported not gated): PAQ beats the best family arm (SnapKV-stale) with min-over-family $\mathrm{LB}_{95} = +0.202$; PAQ − SnapKV-fresh = +0.094 $[+0.069, +0.122]$; PAQ − attn_max = +0.077 $[+0.048, +0.110]$; PAQ − oracle = −0.199 $[-0.239, -0.161]$.

Gold-page recall at b=5% (per cell, LongDocURL / MMLongBench-Doc): PAQ 0.494/0.443; `gold_frequency_static` 0.381/0.373; `paq_stale` 0.137/0.148; ColQwen2-fresh 0.664/0.536.

B The adversarial review trail (anti-strawman record)

This project's positives were only believed after surviving independent cross-family review; two of them did not survive, and we list the retractions because they are part of the evidence:

1. **Pre-freeze design review** (code-faithfulness + methodology, 2026-07-14): folded best-of-family gating, document-cluster bootstrap, the page-routing estimand disclosure, first-M stale aggregation, and power floors *before* any GPU run.

Table 6: Every arm, accuracy at b=5%, per cell (verbatim from `research/paq_verdict.json`). The query-blind family sits at/below random gold-recall — the anti-strawman record: beating it is visibly not the claim.

Group	Arm	LongDocURL	MMLongBench-Doc
ours	PAQ	**0.386**	**0.258**
	attn_max (alt-agg)	0.337	0.152
isolation	paq_stale (commit-once)	0.128	0.095
	gold_frequency_static	0.278	0.154
deployable retrieval	ColQwen2-fresh	0.512	0.328
	ColQwen2-stale	0.216	0.138
	CLIP-fresh	0.155	0.164
reference (not competitors)	SnapKV-fresh	0.298	0.157
	full-context	0.546	0.397
	oracle-evidence (not strict)	0.609	0.431
query-blind controls	SnapKV-stale	0.083	0.074
	H2O	0.045	0.039
	FastV (ctx-only proxy)	0.045	0.031
	LOOK-M (→H2O on images)	0.045	0.039
	KVzip (recv_max proxy)	0.047	0.048
	random	0.068	0.022
	truncation	0.033	0.041

2. **v1 gate retraction.** The first gate (`PAQ_V1_GO`, PAQ beats the eviction family) *passed and was then retracted*: reviewers + an independent reproduction showed a gold-informed static nearly matched PAQ's recall — the family's weakness was a weak-static artifact, and reversibility had not been isolated. The two-comparator isolation gate (Eq. (6)) replaced it and was frozen before the re-run.

3. **Result review of the isolation positive** (2026-07-14): heterogeneity honestly surfaced (screen-scale MMLongBench static contrast −0.017, CI incl. 0 — later resolved at 80 docs); `attn_best`, a derived arm that peeked at gold task score, was dropped.

4. **Final AAAI-readiness review** (2026-07-15): named the missing deployable retrieval baseline as the #1 threat — running it produced the decisive loss (§7.2); forced the `best_static` → `gold_frequency_static` rename and the LB_{95} relabel; demoted the mechanism claim (§7.3); required the independent-extension and missingness analyses (§7.1).

5. **Efficiency red-team** (2026-07-15): rejected efficiency-only framing (§7.5) and predeclared the non-inferiority bars PAQ-T must meet.

C Reproducibility

- **Harness** (frozen): `scripts/paq.py` (v1 sha256 `735eecaa…`, commit `ca66292`; v2 `e9e06aad…`, commit `f7e7a53`; v3 = v2 + confirmatory pin, re-scored at commit `b653e85`). Retrieval arms: `scripts/paq_retrieval_arms.p` (ColQwen2 `vidore/colqwen2-v1.0` + CLIP; incremental merge — the frozen gate math and the retrieval-free rows are untouched). PAQ-T gate: `scripts/paq_t_gate.py`. Figures: `scripts/paq_figs.py` (all numbers read from the verdict JSONs; every figure carries a source stamp `pin·n_rows·verdict`).

- **Pins**: screen `research/paq_pinned_subset.json` (sha256 `b194ed84…`, chmod 444, byte-verified by the harness); confirmatory `research/paq_confirm_pin.json` (80 docs/cell; the screen pin is its prefix).

- **Verdicts**: `research/paq_verdict.json` (gate + budget curves + retrieval secondary + strata), `research/paq_t_gate_verdict.json`; per-query per-arm rows in `research/paq_results.jsonl`; cached retrieval scores in `research/paq_retrieval_scores.json`.
- **Gate + bootstrap + pairing** are unit-tested (`paq.py --selftest`); the belief gate, arm classifications, and strata were pre-registered in the frozen bar.
- **Commands**: gate `python scripts/paq.py --run --cell vision --gate-only`; full contest `--run --cell all`; scoring `--score`; figures `python scripts/paq_figs.py`.

D Timeout recovery note

The confirmatory hit its 10-hour wall-clock limit during final scoring (exit 124) at 903/907 vision queries. Because the harness flushes each row as it completes, the run was recovered by re-scoring the flushed rows on CPU (no GPU re-run, no selection re-computation); the 4 unscored queries are the `mi_phone` rows of §7.1, and the worst-case imputation leaves every gate conclusion unchanged. A separate single-query text cell ("counting") was truncated by the same timeout and is out of scope for the multi-query claims; it was not re-run.

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